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Meeting Abstract

47.2 MCWILLIAMS, S.R.*; PIERCE, B.J.; WITENZELLNER, A.; SPEAKMAN, J.; LANGLOIS, L.; ENGEL, S.; SMITH, S.B.; GUGLIELMO, C.; Univ. Rhode Island, Sacred Heart Univ., Max-Planck-Institute for Ornithology, Univ. Aberdeen, Univ. Western Ontario; srmcwilliams@uri.edu

Dietary fat and exercise performance of migratory birds

Fatty acid composition of fat stores affects exercise performance in a variety of vertebrates although no such studies focus on birds. We tested the hypothesis that birds with more essential n-6 PUFA (e.g., 18:2n-6) comprising their fat stores have enhanced exercise performance during short-term exercise and during 6-hr flights. We fed Red-eyed vireos (*Vireo olivaceus*) and European Starlings (*Sturnis vulgaris*) one of two diets that differed only in the amount of monounsaturated (MUFA) and polyunsaturated (PUFA) fatty acids. As expected, dietary fat composition affected fat composition of birds. Blood metabolites measured prior to and immediately following exercise confirmed that birds were metabolizing endogenous fat during exercise. Resting metabolic rate (2.75 ± 0.32 mlO₂/g/hr) was similar for birds fed either diet. However, during short-term exercise the peak metabolic rate of birds fed more MUFA (28.55 ± 1.47 mlO₂/g/hr) was 25% higher than birds fed more PUFA (21.50 ± 1.76 mlO₂/g/hr). Similarly, during continuous 6-hr flights the energy expenditure of birds fed more MUFA (8.51 ± 0.27 Watts) was 13% higher than birds fed more PUFA (7.5 ± 0.28 Watts). Such whole-animal energetic effects of fatty acid composition of birds suggest that birds during migration can influence the composition of their body fat by selective feeding and this can significantly affect the energetic cost of flights. Supported by NSF (IBN-9983920) and USDA (RIAES-538748).